

Energy Markets - 2on Laboratory Assignment

Analysis of the effect in the electricity market equilibrium of a wind-power plant

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Analysis of the effect in the electricity market equilibrium of a wind-power plant

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Description of the study

Let's consider the market clearing problem defined in the annex. This problem represents a simplified representation of the countrywide wholesale electricity market system (let's say, the day-ahead market, for instance). The bid functions in Table 2 and Table 3 can be considered as the bids of the different agents during a representative working day and the equilibrium point associated to these data represents the current situation of the market. With the help of GAMS codes developed during the course, we want to perform the following two exercises.

Exercise 1. Base test case

First we would like to analyse the base case equilibrium situation corresponding to the data in the annex:

- Find the equilibrium point: nodal prices and traded energy at each bus and time period.
- Describe with full detail the equilibrium point at some bus and time period: aggregated generation and demand functions,
- Describe the power flow and analyse how the power flow at the different transmission lines depends on time. Is there any transmission lines congested? Can you infer any relation between congestion and energy prices?
- Analyse the economical transactions: producers, consumers and merchandising surplus at each time period. Specifically, discuss the interpretation of the merchandising surplus at time periods 3 and 4¹.

Exercise 2. Analysis of the installation of a new wind power plant

We would like to assess the impact in the market equilibrium of the installation of a new wind-power generation plant (WPP). The data for this new generation unit are:

- The WPP is attached to bus number 3.
- The expected energy generation of the WPP is:

$t = 1$	$t = 2$	$t = 3$	$t = 4$	$t = 5$
5MW	10MW	15MW	20MW	10MW

Table 1: expected energy generation of the wind power plant

- The WPP bid all its expected generation at 0€/MWh
- The WPP is not limited neither by ramp limits nor by maximum generation limits².

¹ You can find some clues to interpret the behaviour of the merchandising surplus in these time periods in the paper "Nodal prices and transmission rights: A critical appraisal" by Shmuel S. Oren, Pablo T. Spiller, Pravin Varaiya and Felix Wu (The Electricity Journal, Volume 8, Issue 3, April 1995, Pages 24–35) available at

https://www.researchgate.net/profile/Pablo_Spiller/publication/222972886_Nodal_prices_and_transmission_rights_A_critical_appraisal/file/9c960516d7f5501028.pdf?origin=publication_detail

Use the clearing model *TCMPA* to compare the equilibrium point for this new situation with the equilibrium point of the base test case. Evaluate the impact of this new WPP in:

- a) The nodal prices and traded energy at each bus and time period.
- b) The congestion of the transmission network.
- c) The economical settlement of the market: total social welfare; producer, consumers and merchandising surplus. Does the introduction of the WPP produce some clear benefit, either for the wholesale market system or for any of its participants (producers, consumers, transmission system operator)?

Report

You should elaborate and upload to the intranet a report with the answers to the different sections of Exercise 1 and Exercise 2, taking into account that:

- 1) The same groups of students of the first assignment apply. Any change in the composition of the groups must be communicated previously to the lecturer in charge of the assignment (prof. F.-Javier Heredia, f.javier.heredia@upc.edu).
- 2) The report must follow the same IEEE template used in the first assignment and should include the different parts of the studies proposed in exercises 1 and 2, presented in a comprehensive way following the usual structure of a IEEE paper³:
 - I. Introduction
 - A. Brief description of the problem.
 - B. Objectives of the study and main contributions.
 - C. Structure of the report.
 - II. Description of the problem and mathematical model.
 - III. Numerical results.
 - A. Description of the computational test: data, software, computational resources.
 - B. Results for the base case test (Exercise 1).
 - C. Results for the test case with the WPP (Exercise 2).
 - D. Analysis of the results (comparison of the equilibrium points of the two cases, answer to sections a), b) and c) of Exercise 1
 - IV. Conclusions of the study.
- 3) The deadline for the report is 25th June, 13:00 (1:00 p.m).
- 4) The grading criteria to be applied is :
 - 50% : answers to exercise 1.
 - 30% : answers to exercise 2.
 - 20% : quality of the report.

² That means that the ramp and generation limits constraints of model *TCMPA* should not be applied to this new generation unit. The presence of these constraints in GAMS can be avoided filtering (dollar condition) the definition of constraints **RAMP_D**, **RAMP_U**, **PGMIN** and **PGMAX** (see GAMS User's Guide Chapter 11).

³ An example of how to arrange a IEEE paper can be found at <http://hdl.handle.net/2117/20642>

Annex: base case market data⁴

Generation units	Bus	$P_{G_i}^{max}$ [MW]	$P_{G_i}^{min}$ [MW]	$R_{G_i}^{up}$ [MW]	$R_{G_i}^{down}$ [MW]	$P_{G_i}^0$ [MW]	$P_{G_{ikt}}^b [MW] \forall t$		$\lambda_{G_{ikt}}^b \left[\frac{\text{€}}{\text{MWh}} \right] \forall t$	
							$k = 1$	$k = 2$	$k = 1$	$k = 2$
1	1	300	30	100	100	150	100	200	10	30
2	1	275	28	30	30	28	50	25	40	65
3	2	100	10	25	20	50	30	70	40	59
4	2	50	5	10	10	25	30	20	40	67
5	4	200	20	50	50	100	100	0	30	30

Table 2: data for the generation units.

Demand units		Bus	$P_{D_{ikt}}^b [MW]$			$\lambda_{D_{ikt}}^b [€/MWh]$		
			$k = 1$	$k = 2$	$k = 3$	$k = 1$	$k = 2$	$k = 3$
$t = 1$	$i = 1$	1	1	0	0	100	0	0
	$i = 2$	2	100	100	40	100	80	75
	$i = 3$	3	80	50	30	100	65	35
$t = 2$	$i = 1$	1	1	0	0	60	0	0
	$i = 2$	2	80	80	32	60	48	45
	$i = 3$	3	64	40	24	60	39	21
$t = 3$	$i = 1$	1	2	0	0	120	0	0
	$i = 2$	2	170	170	68	120	96	90
	$i = 3$	3	136	85	51	120	78	42
$t = 4$	$i = 1$	1	2	0	0	140	0	0
	$i = 2$	2	130	130	52	140	112	105
	$i = 3$	3	104	65	39	140	91	49
$t = 5$	$i = 1$	1	2	0	0	130	0	0
	$i = 2$	2	140	140	56	130	104	98
	$i = 3$	3	112	70	42	130	85	46

Table 3: data for the demand units.

Transmission lines ($sb = 100MW$)			
$k =$	$l =$	$x_{kl} [p.u]$	$S_{kl}^{max} [MW]$
1	2	0.058	150
1	3	0.04	150
2	3	0.08	150
2	4	0.08	150
3	4	0.12	150

Table 4: transmission network

⁴ Adapted from the example in section 6.2.2. of the book “Regulation of the Power Sector”, I.J. Pérez-Arriaga (Editor). Springer (2013).